Supplementary Material for: Evidence-Free Decision Points in Biomarker-Driven Metastatic Cancer Guidelines

Hsieh-Ting Lin[1,*]

[1]Department of Hematology & Medical Oncology, Koo Foundation Sun Yat-Sen Cancer Center, 125 Lide Rd., Beitou Dist., Taipei 11259, Taiwan. ORCID iD: 0009-0002-3974-4528.
[*]Correspondence: `htlin222@kfsyscc.org`.

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This document supplements the main manuscript *Evidence-Free Decision Points in Biomarker-Driven Metastatic Cancer Guidelines: A Pre-Registered Multi-Tumor Audit of ESMO, ASCO, and NCCN Decision Trees in HR+/HER2- Breast Cancer and EGFR/ALK Non-Small-Cell Lung Cancer* (Paper A; tagged release `v3.0.0`; Zenodo DOI `10.5281/zenodo.20250848`).

The four tables provided are:

- **Table S1.** PRISMA-2020 reporting items modified for graph-theoretic guideline-trial concordance analysis (with graph-encoding addendum).
- **Table S2.** Representative per-trial structured extractions (10 of 259 pivotal trials).
- **Table S3.** Full 49-node ESMO/ASCO/NCCN unified decision tree with strict-tolerance concordance status.
- **Table S4.** Adjudication rules log consolidating the deterministic rules applied when LLM annotators disagreed.

S1: PRISMA-2020 reporting checklist (modified)

Table 1. PRISMA-2020 reporting items modified for graph-theoretic guideline-trial concordance analysis. Items reflect the systematic-search and structured-extraction core of PRISMA-2020 (Page *et al.* 2021), with a graph-encoding addendum at the end specific to this study type.

#	Item	Location in manuscript / repository
1	Title identifies study as systematic with concordance computation	Title (both papers)
2	Structured abstract with Purpose / Methods / Results / Conclusion	Abstract section
3	Rationale: established gap in literature	Introduction
4	Objectives stated as testable pre-specified question (EFDPR vs $p_0 = 0.25$)	Introduction final paragraph; pre-registration commits `4b5bf1a`
5	Eligibility criteria (population, intervention, comparator, outcomes)	Methods §**??**, `prereg-v3.md`
6	Information sources: ClinicalTrials.gov v2 API; ESMO/ASCO/NCCN guideline documents	Methods; `analysis/v3_01_systematic_search_nsclc.py`
7	Search query reproduced verbatim	`analysis/v3_01_systematic_search_nsclc.py`
8	Selection process with two-pass filter and audit trail	`analysis/v3_02_filter_nsclc.py`; `data/processed/v3_nsclc_filter_log.json`
9	Data collection: dual-LLM-annotator pipeline (Claude 4.7 + Codex/GPT-5) with adjudication	Methods §dual-annotator; `extraction_protocol.md` & `extraction_protocol_nsclc.md`
10	Data items: frozen JSON schema (state, biomarker, drug class, line of therapy)	`extraction_protocol*.md` "Output schema"
11	Bias assessment: pre-adjudication Cohen's κ and PABAK with documented gate failures	Methods §inter-rater; `data/results/v3_nsclc_kappa.json`
12	Synthesis methods: graph-theoretic concordance with strict / ESCAT / liberal biomarker-tolerance grid	Methods §framework
13	Reporting bias assessment: pre-registration of primary test prevents outcome-dependent analysis-choice	`docs/prereg-v3.md`
14	Certainty assessment: one-sided exact binomial with Clopper-Pearson exact CI + bootstrap sensitivity	Methods §statistical inference
15	Search results: 702 NSCLC candidates $\rightarrow$ 178 pivotal trials (5 supplementary additions after round-1 audit); mBC corpus 81 trials from v2.0.0	Results §corpus; `data/processed/v3_nsclc_filter_log.json`
16	Study characteristics summary (Tables S2–S3)	Supplement
17	Risk of bias of individual studies: not applicable at the trial-extraction level; bias of the extraction process is assessed via dual-LLM-annotator κ/PABAK	Methods
18	Results: EFDPR 0.39, P=0.023 (pooled strict); see Figure 2 per-node breakdown	Results; Figure 2
19	Results of syntheses: ESCAT/liberal sensitivity (P=0.084); tumour-stratified analyses	Results §sensitivity
20	Reporting bias: pre-registration trail (commits `085ae54`, `079b540`, `4b5bf1a`) limits researcher degrees of freedom	Methods §pre-registration
21	Certainty of evidence: one-node-deep rejection disclosed; BH-q across 15-cell tolerance grid disclosed	Discussion §caveats
22	Discussion: 3 caveats + 3 positives	Discussion

S2: Representative per-trial structured extractions

Table 2. Representative per-trial structured extractions (10 of the 259 pivotal trials in the v3.0.0 corpus). Full dual-annotator outputs released at `data/processed/v2_extraction_final.json` and `nsclc_full.json`.

NCT ID	Trial name	Pre-treatment state	Biomarker / subgroup	Drug class
NCT01958021	MONALEESA-2	first-line	HR+/HER2-	CDK4/6i + AI
NCT02278120	MONALEESA-7	first-line+pre-menopa…	HR+/HER2-	CDK4/6i + endocrine (pre-me…
NCT01942135	PALOMA-3	post-endo	HR+/HER2-	CDK4/6i + fulvestrant
NCT03778931	EMERALD	post-endo+post-CDK46i	HR+/HER2-/ESR1mut	SERD oral
NCT04494425	DESTINY-Breast06	post-endo+post-CDK46i	HR+/HER2-/HER2-low	HER2-ADC (T-DXd)
NCT01154140	PROFILE 1014	first-line	ALK-rearranged/NSCLC	ALK TKI 1st-gen (crizotinib)
NCT02296125	FLAURA	first-line	EGFR-mut/EGFR-ex19del/EGFR-…	EGFR TKI 3rd-gen (osimertin…
NCT04487080	MARIPOSA	first-line	EGFR-mut/EGFR-ex19del/EGFR-…	EGFR TKI 3rd-gen + bispecif…
NCT02151981	AURA3	post-EGFRTKI	EGFR-mut/EGFR-T790M/NSCLC	EGFR TKI 3rd-gen (osimertin…
NCT03521154	LAURA	first-line	EGFR-mut/EGFR-ex19del/EGFR-…	EGFR TKI 3rd-gen (osimertin…

S3: Full 49-node guideline decision tree with concordance status

Table 3. Full 49-node ESMO/ASCO/NCCN unified decision tree with strict-tolerance concordance status. “Concordant” = at least one trial edge satisfies state-superset, biomarker-superset, drug-class, and temporal-precedence criteria. “Evidence-free” = no such trial edge. Concordant trial counts in last column are derived from `data/results/v3_pooled_efdpr.json`.

Node	Tumor	State	Biomarker	Recommended class(es)	Year	Concordance
G1	mBC	first-line	HR+/HER2-	CDK4/6i + AI	2018	concordant (n=4)
G2	mBC	first-line+pre-me…	HR+/HER2-	CDK4/6i + endocrine (pre-menopausa…	2019	concordant (n=1)
G3	mBC	post-endo	HR+/HER2-	CDK4/6i + fulvestrant	2016	concordant (n=1)
G4	mBC	post-endo	HR+/HER2-/PIK3CAmut	PI3Ki + fulvestrant	2020	concordant (n=2)
G5	mBC	post-CDK46i	HR+/HER2-/ESR1mut	SERD oral	2023	concordant (n=1)
G6	mBC	post-endo	HR+/HER2-/AKTpath	AKTi + fulvestrant	2024	concordant (n=1)
G7	mBC	post-CDK46i	HR+/HER2-/PIK3CAmut	PI3Ki + fulvestrant	2022	concordant (n=1)
G8	mBC	post-CDK46i	HR+/HER2-/HER2-low	HER2-ADC (T-DXd)	2024	concordant (n=1)
G9	mBC	post-CDK46i	HR+/HER2-/no-actionable	everolimus + exemestane; chemother…	2022	**evidence-free**
G10	mBC	post-CDK46i	HR+/HER2-	CDK4/6i + fulvestrant (post-CDK4/6…	2024	concordant (n=2)
G11	mBC	post-endo	HR+/HER2-/HER2-low	HER2-ADC (T-DXd)	2024	concordant (n=1)
G12	mBC	post-chemo	HR+/HER2-/HER2-low	HER2-ADC (T-DXd)	2022	**evidence-free**
G13	mBC	post-CDK46i+post-…	HR+/HER2-	everolimus + exemestane	2018	**evidence-free**
G14	mBC	post-CDK46i+post-…	HR+/HER2-	Sacituzumab govitecan	2023	**evidence-free**
G15	mBC	post-endo	HR+/HER2-/PIK3CAmut	PI3Ki triplet (inavolisib + CDK4/6…	2024	**evidence-free**
G16	mBC	metastatic	gBRCAmut/HER2-	PARPi (olaparib); PARPi (talazopar…	2018	concordant (n=1)
G17	mBC	first-line+viscer…	HR+/HER2-	chemotherapy single agent	2022	**evidence-free**
G19	mBC	post-endo	HR+/HER2-/ESR1mut	SERD oral + CDK4/6i	2024	concordant (n=1)
G20	mBC	post-endo+post-CD…	HR+/HER2-	TROP2-ADC (sacituzumab govitecan)	2023	concordant (n=1)
G21	mBC	post-chemo	HR+/HER2-/HER2-low	TROP2-ADC (datopotamab deruxtecan)…	2025	**evidence-free**
G22	mBC	first-line+indole…	HR+/HER2-	endocrine therapy alone	2020	**evidence-free**
G23	mBC	post-endo+post-CD…	HR+/HER2-	TROP2-ADC (sacituzumab govitecan)	2023	concordant (n=1)
G24	mBC	first-line+indole…	HR+/HER2-	endocrine therapy alone	2018	**evidence-free**
G25	mBC	post-CDK46i+post-…	HR+/HER2-	chemotherapy single agent	2022	**evidence-free**
G26	mBC	post-CDK46i	HR+/HER2-/AKTpath	AKTi + fulvestrant	2024	concordant (n=1)
N1	NSCLC	first-line	EGFR-mut/NSCLC	EGFR TKI 3rd-gen (osimertinib)	2018	concordant (n=4)
N2	NSCLC	first-line	EGFR-mut/NSCLC	EGFR TKI 3rd-gen + chemotherapy	2024	concordant (n=1)
N3	NSCLC	first-line	EGFR-mut/NSCLC	EGFR TKI 3rd-gen + bis-…	2024	concordant …

S4: Adjudication rules log

Table 4. Adjudication rules log consolidating the deterministic decision rules used when the two LLM annotators disagreed on a structured-extraction field. Full per-trial annotator outputs and the JSON-level adjudicated record are released in `data/processed/v2_extraction_final.json` (mBC) and `nsclc_full.json` (NSCLC). Rules were pre-specified in `analysis/extraction_protocol.md` (committed `4b5bf1a`, before any outcome-touching analysis).

Field	Adjudication rule
mBC HR+/HER2- pipeline rules	
`post_endo` vs `excluded.post_endo_metastatic_any`	"Must have progressed on prior endocrine therapy in the metastatic setting" ⇒ `post_endo: true`. "No prior systemic therapy in the metastatic setting" ⇒ `excluded.post_endo_metastatic_any: true`. Silent ⇒ both `null`.
`post_cdk46i`	"Must have received prior CDK4/6i" ⇒ `true`. "Prior CDK4/6i is permitted" ⇒ `null` (allowed, not required). "No prior CDK4/6i" or "CDK4/6i-naive" ⇒ `false` AND `excluded.post_cdk46i: true`.
`biomarker.hr_pos`	Eligibility requires HR+ or ER+ ⇒ `true`. Eligibility enrols all-comers ⇒ `null` if HR+ is a stratification; `false` if explicitly HR-.
`biomarker.her2_neg` vs `her2_low`	"HER2-negative" (IHC 0/1+/2+ ISH-) ⇒ `her2_neg: true`, `her2_low: null`. Specifically requires HER2-low (IHC 1+ or 2+/ISH-) ⇒ both `her2_neg: true` AND `her2_low: true`. Requires HER2+ (IHC 3+ or ISH-amplified) ⇒ `her2_neg: false`; trial flagged out-of-scope.
`biomarker.pik3camut` & `biomarker.esr1mut` & `biomarker.akt_path`	Extracted only when explicitly required by trial eligibility; `null` otherwise. `akt_path` adjudication kappa failed gate (PABAK pass) — disclosed in Methods.
NSCLC EGFR/ALK pipeline rules	
`tumor_type`	"NSCLC-EGFR" if eligibility requires EGFR mutation; "NSCLC-ALK" if ALK rearrangement; "NSCLC-both" if multi-cohort; "NSCLC-other" ⇒ trial flagged out-of-scope (HER2-mut, KRAS G12C, ROS1, RET, MET, NTRK, BRAF, driver-negative).
mBC-specific fields on NSCLC trials	All `null` or `false`; never inferred.
`post_egfr_tki` vs `post_osimertinib`	Post-1st/2nd-gen TKI progression ⇒ `post_egfr_tki: true`. Post-osimertinib (3rd-gen) progression ⇒ both `post_egfr_tki: true` AND `post_osimertinib: true`. Silent ⇒ `null`.
`alk_resistance_mut`	If trial enrols patients with specific ALK resistance mutations (e.g., G1202R) ⇒ `true`; else `null`.
`drug_class`	Prefer most specific canonical string. For combination arms, use the experimental-arm class.
`guideline_target_node`	Format `<state>|<biomarker>|NSCLC`, e.g. `post-osimertinib|EGFR-mut|NSCLC`.
Cross-pipeline rules	
Out-of-scope handling	Out-of-scope trials are retained in the extraction record with `confidence: "low"` and `notes: "Out of scope: <reason>"` for transparency; they do not enter the trial-DAG.
Tie-breaking	When annotators disagreed on a Boolean field that flips trial inclusion, the adjudication outcome is determined by re-reading the verbatim eligibility text per the rules above. The two annotator outputs (pre-adjudication) and the adjudicated record (post-adjudication) are released side-by-side in `v2_extraction_final.json`.
Round-1 audit corrections	Four NCT-misidentifications (AURA3, HERTHENA-Lung01, TROPION-Lung01, FLAURA-CNS) and six encoding bugs were caught during round-1 internal review and corrected with hand-curated records before final analysis. The audit trail is in commit history (`4ff3cdd`, `dadf797`, `123a461`).